

AI Proposal & Report Writer — MVP

A FastAPI application that uses GPT-4 to generate professional proposals and reports, exported as branded PDFs.

Project Structure

```
app/  
├─ main.py           # FastAPI app & OpenAI integration  
├─ pdf_generator.py  # ReportLab PDF builder  
├─ requirements.txt  
├─ static/  
│   └─ index.html    # Single-page frontend  
└─ output/           # Generated PDFs saved here (auto-created)
```

Setup & Run

1. Prerequisites

- Python 3.10 or higher
- An OpenAI API key

2. Create a virtual environment

```
bash  
  
cd app  
python -m venv venv  
  
# Activate (macOS/Linux)  
source venv/bin/activate  
  
# Activate (Windows)  
venv\Scripts\activate
```

3. Install dependencies

```
bash  
  
pip install -r requirements.txt
```

4. Set your OpenAI API key

```
bash

# macOS/Linux
export OPENAI_API_KEY="sk-..."

# Windows (Command Prompt)
set OPENAI_API_KEY=sk-...

# Windows (PowerShell)
$env:OPENAI_API_KEY="sk-..."
```

5. Run the server

```
bash

uvicorn main:app --reload --port 8000
```

The app will be available at <http://localhost:8000>

Using the App

Browser (recommended)

Open <http://localhost:8000> — fill in the form and click **Generate PDF**. The file downloads automatically.

API (curl)

```
bash

curl -X POST http://localhost:8000/generate-report \
  -H "Content-Type: application/json" \
  -d '{
    "client_name": "TechStart Inc.",
    "project_description": "Build a customer analytics dashboard with real-time data",
    "report_type": "proposal",
    "template_id": "default"
  }' \
  --output proposal.pdf
```

Parameters:

Field	Type	Required	Values
<code>client_name</code>	string	✓	Any name
<code>project_description</code>	string	✓	Free text
<code>report_type</code>	string	✓	"proposal" or "report"
<code>template_id</code>	string	✗	"default" (only option for now)

Output

- Generated PDFs are saved to the `output/` folder automatically.
- Filenames follow the pattern: `proposal_techstart_inc_a1b2c3.pdf`

Template Customization

Edit the `TEMPLATES` dict in `pdf_generator.py` to change:

- **Company name and logo text**
- **Primary / accent colors** (hex codes)
- **Footer contact details**

AI Model

The app tries **GPT-4o** first, then falls back to **GPT-3.5-turbo** automatically if the first model is unavailable or quota-limited.

Interactive API Docs

FastAPI auto-generates docs at:

- Swagger UI: <http://localhost:8000/docs>
- ReDoc: <http://localhost:8000/redoc>